

Fractions:

Half & Quarter

Grade 4 Mathematics Lesson

Learning Objective: Understand and use $\frac{1}{2}$ and $\frac{1}{4}$ in everyday life

$\frac{1}{2}$
Half

$\frac{1}{4}$
Quarter

$\frac{3}{4}$
Adding

Subject	Mathematics – Number Sense
Grade	Grade 4
Topic	Fractions: Half ($\frac{1}{2}$) and Quarter ($\frac{1}{4}$)
Duration	45 minutes

1. What Are Fractions?

Definition:

Fractions show **parts of a whole**. When we divide something into **equal pieces**, each piece is a fraction of the whole thing!

Key Vocabulary

Term	Meaning	Example
Fraction	A part of a whole	$1/2$, $1/4$
Numerator	Top number — how many parts we have	1 in $1/2$
Denominator	Bottom number — total equal parts	2 in $1/2$
Equal parts	All pieces are exactly the same size	Both halves of a pizza

2. Understanding Half ($\frac{1}{2}$)

What is Half?

Half means dividing something into **2 equal parts**. Both pieces must be exactly the same size!

- One half is written as $1/2$
- The top number (1) shows how many parts we have
- The bottom number (2) shows the total equal parts
- Both halves together make **one whole**!

Remember!

When something is split in half:

- ✓ 2 equal parts are created
- ✓ Each part = $1/2$
- ✓ $1/2 + 1/2 = 1$ whole

$1/2$

Real-Life Examples of Half

■ Pizza	Cut a pizza down the middle → 2 equal halves. Eat one half, your friend gets the other!
■ Orange	Cut an orange in the middle → two equal halves, each with the same juicy segments.
■ Chocolate	Break a chocolate bar in half → you and a friend each get the same amount. Fair sharing!
■ Rectangle	Draw a line through the middle of a rectangle → each side is exactly $1/2$.

3. Understanding Quarter ($\frac{1}{4}$)

What is a Quarter?

A quarter means dividing something into **4 equal parts**. When we take one of those four parts, we call it one quarter.

- Each part is the same size
- One quarter = 1 out of 4 parts
- Written as **$\frac{1}{4}$**
- Four quarters make **one whole**!

Remember!

When something is divided into quarters:

- ✓ 4 equal parts are created
- ✓ Each part = $\frac{1}{4}$
- ✓ $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$ whole

$\frac{1}{4}$

Real-Life Examples of Quarter

■ Pizza	Share a pizza with 3 friends → each person gets 1 quarter = $\frac{1}{4}$ of the pizza.
■ Cake	A round cake cut into 4 slices → one slice is $\frac{1}{4}$ of the whole cake.
■ Watermelon	A big watermelon cut into 4 pieces → each piece is one quarter.
■ Chocolate	A chocolate bar with 4 equal blocks → each block is $\frac{1}{4}$ of the bar.

4. Comparing Half and Quarter

Half ($\frac{1}{2}$)

- Divided into **2** equal parts
- Each part is **larger**
- $\frac{1}{2}$ means 1 out of 2 parts
- Example: Half a pizza = 2 big slices

1 /

Quarter ($\frac{1}{4}$)

- Divided into **4** equal parts
- Each part is **smaller**
- $\frac{1}{4}$ means 1 out of 4 parts
- Example: Quarter pizza = 4 small slices

1 /

■ Key Insight

When the denominator (bottom number) is **larger**, the pieces are **smaller**. So $\frac{1}{4}$ is smaller than $\frac{1}{2}$ because the whole is divided into more pieces. Think: sharing a pizza between 4 people gives everyone less than sharing between 2 people!

5. Adding Fractions: $\frac{1}{2} + \frac{1}{4}$

Step 1	Make the denominators equal	$\frac{1}{2} = \frac{2}{4}$ (multiply top and bottom by 2)
Step 2	Add the numerators	$\frac{2}{4} + \frac{1}{4}$ (the bottom stays the same)
Step 3	Write the answer	$\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ ← three quarters!

Result: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

6. Practice Problems

Question 1: Shade **half** of a circle.

Hint: Draw a line through the middle and colour one side.

Question 2: Colour **one quarter** of the rectangle below.

Hint: Divide it into 4 equal parts, then colour just one part.

Question 3: What is $\frac{1}{2} + \frac{1}{4}$? Show your working.

Hint: Remember to make the denominators equal first!

Question 4: Which is bigger: $\frac{1}{2}$ or $\frac{1}{4}$? Explain why.

Hint: Think about how many equal parts each fraction has.

7. Answers & Explanations

Q1	Shade half of a circle	Draw a straight line through the centre of the circle. Colour one of the two equal halves.
Q2	Colour one quarter	Draw two lines through the rectangle to make 4 equal parts. Colour exactly 1 of the 4 parts.
Q3	$\frac{1}{2} + \frac{1}{4}$	Convert $\frac{1}{2} \rightarrow \frac{2}{4}$, then add: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$. The answer is three quarters!
Q4	Which is bigger?	$\frac{1}{2}$ is bigger than $\frac{1}{4}$. Half splits into only 2 parts, so each piece is larger.

8. Fun Quiz Time! ■

Question 1: True or False: Half means dividing into 2 equal parts.

a) True b) False

Question 2: Which is bigger: $\frac{1}{2}$ or $\frac{1}{4}$?

a) $\frac{1}{2}$ b) $\frac{1}{4}$ c) They are equal

Question 3: If you eat $\frac{1}{4}$ of a pizza, how many equal pieces are left?

a) 1 piece b) 2 pieces c) 3 pieces

Question 4: What is the denominator in the fraction $\frac{1}{2}$?

a) 1 b) 2 c) 3

Question 5: How many quarters make one whole?

a) 2 b) 3 c) 4

Quiz Answers: 1. True | 2. a) $\frac{1}{2}$ | 3. c) 3 pieces | 4. b) 2 | 5. c) 4

9. Lesson Summary

■	Fractions show parts of a whole when divided into equal pieces.
■	Half ($\frac{1}{2}$) = dividing into 2 equal parts. Both halves together = 1 whole.
■	Quarter ($\frac{1}{4}$) = dividing into 4 equal parts. Four quarters together = 1 whole.
■	$\frac{1}{2}$ is bigger than $\frac{1}{4}$ — fewer parts means each part is larger.
■	To add $\frac{1}{2} + \frac{1}{4}$: convert to $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$.
■	Fractions are used every day — cooking, sharing food, measuring!

Next Steps ■

- Find halves and quarters around your home today!
- Practice drawing shapes and dividing them into equal parts.
- Try $\frac{1}{4} + \frac{1}{4}$ — can you work it out?
- Ask your teacher if you need any help!